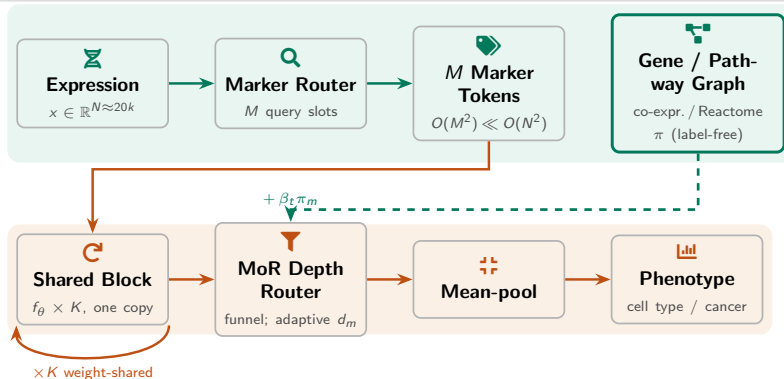


Problem & design: a biology-informed recursive transformer

Problem statement

Classify a cell/sample from $x \in \mathbb{R}^N$ ($N \approx 20k$ genes) *accurately & cheaply*. Two costs are our targets: **(T) token cost** — attention is $O(N^2)$ in genes; **(F) FLOP/depth cost** — a K -layer stack is $O(K)$ FLOPs and $O(K)$ params, spent uniformly on every token. **Goal: (T) $O(N^2) \rightarrow O(M^2)$, $M \ll N$; (F) $O(K) \rightarrow$ adaptive per-token depth at constant params** — guided by biology.



A marker selection: $N \rightarrow M$ tokens, $O(M^2)$ attention • **B** one shared block $\times K$ + **MoR** adaptive depth • **prior** π nudges **hubs** deeper (label-free)

Results: efficiency wins, accuracy is a wash (honest negatives)

Token reduction

macro-F1 vs # marker tokens M

	$M=16$	$M=64$	$M=128$	$M=256$
Tabula Muris	35.4	64.6	71.3	75.3
Prototype	62.5	90.2	94.8	97.2
Pancreas	43.9	53.0	58.4	57.7
Prostate	74.1	78.5	77.0	78.2

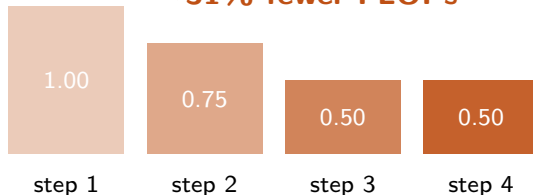
Vanilla vs Recursive vs MoR @ $K^*=4$

pancreas	Vanilla	Recursive	MoR
Params (unique)	299,136	74,784	75,168
FLOPs/sample	62.9 M	62.9 M	38.9 M
Compute saved	—	—	38%
macro-F1	54.1 ± 0.8	57.3 ± 5.8	54.2 ± 0.9
accuracy	92.4	91.8	91.1

Equal accuracy: MoR = $4 \times$ fewer params than Vanilla + 38% fewer FLOPs

Adaptive depth $\Rightarrow$ compute funnel

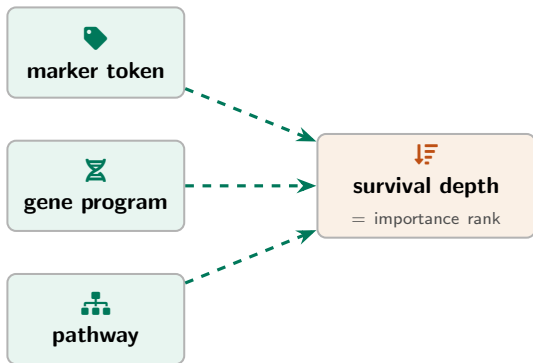
mean depth $2.75/4$
 $\sim 31\%$ fewer FLOPs



Verdicts (multi-seed, statistical)

- + Adaptive depth: $\sim 31\%$ compute saved at matched accuracy & params ($p < 0.001$).
- Biological prior vs random-graph: $\Delta F1 = +0.28$, $p = 0.871$, **0/10 significant**.
- Adaptive vs vanilla accuracy & calibration: **no consistent gain**.

Biologically meaningful token reduction — and where to push next



Each token $\approx$ a learned **signature / gene program** — a deployable **marker panel**, not a black box. A token's **recursion depth** is a compute-based **importance score**.

Negatives $\Rightarrow$ directions

- **Stronger injection:** prior is an additive, annealed-away bias $\rightarrow$ use **FiLM/gating** so centrality *modulates* routing per cell.
- **Better graph:** swap co-expression centrality for **directed GRNs / perturbation graphs** (causal, not correlational).
- **Right granularity:** route **pathway tokens** $\Rightarrow$ "depth per pathway" = compute a biological process needs.
- **Validate:** check survival-depth rankings against **known lineage / disease markers**.

Take-away: token reduction is the biologically meaningful lever that *works*; the bio prior is a promising direction, reported honestly.